

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-7:

Aim: Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution

PROCEDURE :

1. Compilation :

```
venkatsaikrishna5@VSK-HP:~$ mkdir skill7
venkatsaikrishna5@VSK-HP:~$ cd skill7
venkatsaikrishna5@VSK-HP:~/skill7$ nano path_search.c
venkatsaikrishna5@VSK-HP:~/skill7$ gcc path_search.c -o path_search
venkatsaikrishna5@VSK-HP:~/skill7$ ./path_search ls
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH/:/mnt/c/Program Files (x86)/pcsuite/:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/:/mnt/c/Users/vsktu/.local/bin:/mnt/c/Users/vsktu/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/vsktu/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/vsktu/Downloads/jdk-21-windows-x64_bin.exe:/mnt/c/Users/vsktu/AppData/Roaming/npm:/snap/bin

Command found: /usr/bin/ls
Parent process waiting for child...

Child process started. PID = 1804
path_search path_search.c
Child terminated with exit status: 0
Parent process completed.
```

2. Valid command:

```
venkatsaikrishna5@VSK-HP:~/skill7$ ./path_search pwd
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/u
sr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files
/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Co
mmon Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/
windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/Wi
ndowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH/:/mnt/c/P
rogram Files (x86)/pcsuite/:/mnt/c/Program Files/nodejs/:/mnt/c
/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8
.0/bin/:/mnt/c/Users/vsktu/.local/bin:/mnt/c/Users/vsktu/AppDat
a/Local/Microsoft/WindowsApps:/mnt/c/Users/vsktu/AppData/Local/
Programs/Microsoft VS Code/bin:/mnt/c/Users/vsktu/Downloads/jdk
-21_windows-x64_bin.exe:/mnt/c/Users/vsktu/AppData/Roaming/npm:
/snap/bin

Command found: /usr/bin/pwd
Parent process waiting for child...

Child process started. PID = 1816
/home/venkatsaikrishna5/skill7
Child terminated with exit status: 0
Parent process completed.
```

3. Missing command:

```
venkatsaikrishna5@VSK-HP:~/skill7$ ./path_search hello123
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/u
sr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files
/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/
windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/Wi
ndowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH:/mnt/c/P
rogram Files (x86)/pcsuite:/mnt/c/Program Files/nodejs:/mnt/c/
/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8
.0/bin:/mnt/c/Users/vsktu/.local/bin:/mnt/c/Users/vsktu/AppDat
a/Local/Microsoft/WindowsApps:/mnt/c/Users/vsktu/AppData/Local/
Programs/Microsoft VS Code/bin:/mnt/c/Users/vsktu/Downloads/jdk
-21_windows-x64_bin.exe:/mnt/c/Users/vsktu/AppData/Roaming/npm:
/snap/bin

Command 'hello123' not found in PATH.
venkatsaikrishna5@VSK-HP:~/skill7$ ./path_search cat
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/u
sr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files
/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/C
ommon Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/
windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/Wi
ndowsPowerShell/v1.0/:/mnt/c/windows/System32/OpenSSH:/mnt/c/P
rogram Files (x86)/pcsuite:/mnt/c/Program Files/nodejs:/mnt/c/
/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8
.0/bin:/mnt/c/Users/vsktu/.local/bin:/mnt/c/Users/vsktu/AppDat
a/Local/Microsoft/WindowsApps:/mnt/c/Users/vsktu/AppData/Local/
Programs/Microsoft VS Code/bin:/mnt/c/Users/vsktu/Downloads/jdk
-21_windows-x64_bin.exe:/mnt/c/Users/vsktu/AppData/Roaming/npm:
/snap/bin

Command found: /usr/bin/cat
Parent process waiting for child...

Child process started. PID = 1843
```

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/stat.h>

int main(int argc, char *argv[])
{
    if (argc != 2)
    {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    char *path_copy = strdup(path);
    char *dir = strtok(path_copy, ":");

    int found = 0;

    while (dir != NULL)
    {
        char fullpath[1024];

        snprintf(fullpath, sizeof(fullpath), "%s/%s", dir, argv[1])
```

```

    struct stat st;

    if (stat(fullpath, &st) == 0)
    {
        if (access(fullpath, X_OK) == 0)
        {
            printf("Command found: %s\n", fullpath);
            found = 1;
            break;
        }
        else
        {
            printf("Found but not executable: %s\n", fullp>
        }
    }

    dir = strtok(NULL, ":");
}

free(path_copy);

if (!found)
{
    printf("Command '%s' not found in PATH.\n", argv[1]);
    return 1;
}

pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    return 1;
}

if (pid == 0)

```

```

if (pid == 0)
{
    printf("\nChild process started. PID = %d\n", getpid());

    execlp(argv[1], argv[1], (char *)NULL);

    perror("exec failed");
    exit(1);
}
else
{
    int status;

    printf("Parent process waiting for child...\n");

    waitpid(pid, &status, 0);

    if (WIFEXITED(status))
    {
        printf("Child terminated with exit status: %d\n",
               WEXITSTATUS(status));
    }

    printf("Parent process completed.\n");
}

return 0;
}

```

Observation :

- The PATH environment variable was retrieved successfully.
- The PATH was divided into separate directories and searched.
- The required executable command was located successfully.
- Execute permission was verified using access().
- A child process was created using fork().
- The parent process waited for the child using waitpid().
- Missing commands were detected and an error message was displayed.

Result :

The program successfully implemented **process synchronization using waitpid()** and **command resolution using the PATH variable**. It correctly searched for executables, verified permissions, executed valid commands, monitored the child process, and handled missing commands successfully.